Mechanistic Modeling

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SNU-668

Theoretical Framework

Objective

Simulate chromosomal evolution in **SNU-668 Control** populations and estimate the **optimal missegregation probability** p that best reproduces the **late-phase** karyotype. Assess **robustness** to population size and perform **empirical hypothesis tests** of neutrality.

Modeling Assumptions

We model karyotype evolution as a **discrete-time stochastic process** (neutral evolution) with symmetric gain/loss during division:

- Symmetric transitions: $p_{\text{gain}} = p_{\text{loss}} = p$.
- Missegregation occurs at cell division; otherwise a perfect copy is produced.
- Copy number is bounded below by 0 (no negative values).
- Initial population is sampled from **early Control** single-cell karyotypes.
- Divisions accumulate across passages.

Simulation Design

Let
- $C_{i,j}^{(t)}$: copy number of chromosome j in cell i at time t,
- p: per-division missegregation probability,
- N: population size (number of cells simulated),
- P: number of passages,
- d: divisions per cell per passage (empirically $d = 7$).

At each step $t \in \{1, \dots, S\}$ with $S = P \times N \times d$:

1. Sample a parent cell i and a target daughter j.

2. With probability p (missegregation), choose chromosomes c_1, c_2 uniformly and set

$$C_{i,c_1}^{(t+1)} = C_{i,c_1}^{(t)} + 1, \qquad C_{j,c_2}^{(t+1)} = \max\{0,\, C_{i,c_2}^{(t)} - 1\}.$$

   If the loss would create a negative value, abort the event.

3. With probability $1-p$, perform a perfect copy: $C_{j,\cdot}^{(t+1)} = C_{i,\cdot}^{(t)}$.

After S steps, compute the **mean simulated profile** $\bar{C}^{(S)}$ across cells.

Parameter Optimization (Log-Grid Search)

We select p^* that minimizes the Manhattan distance between the simulated average and the observed **late Control** profile:

$$D(p) = \left\| \bar{C}^{(S)}(p) - \bar{C}^{\mathrm{late}} \right\|_1 .$$

Procedure 1. Evaluate 50 log-spaced values $p \in [10^{-8},\, 10^{-1}]$.
2. For each p: simulate 10 passages from **early Control**, compute $D(p)$.
3. Choose $p^* = \arg\min_p D(p)$.
4. Run replicate simulations (e.g., $R = 100$) at p^* to quantify variability.

Replicates and Variability Metrics

For R replicate simulations at p^*:

- **External fit**:

$$D_{\mathrm{sim,late}}^{(r)} = \left\| \bar{C}_{(r)}^{(S)} - \bar{C}^{\mathrm{late}} \right\|_1 .$$

- **Internal variability** (reproducibility):

$$D_{\mathrm{sim,sim}}^{(r,r')} = \left\| \bar{C}_{(r)}^{(S)} - \bar{C}_{(r')}^{(S)} \right\|_1 .$$

Summarize means, standard errors, and quantiles of both metrics.

Sensitivity Analysis to Population Size

To test robustness to the number of simulated cells, repeat the entire simulation at p^* for

$$N \in \{100,\ 1{,}000,\ 10{,}000,\ 100{,}000\}.$$

For each N:

1. Run R replicates (e.g., $R = 20$).
2. Compute and report
   $\overline{D}_{\mathrm{sim,late}}(N)$ and its 95% CI,
   $\overline{D}_{\mathrm{sim,sim}}(N)$ and its 95% CI.
3. A near-constant $\overline{D}_{\mathrm{sim,late}}(N)$ and $\overline{D}_{\mathrm{sim,sim}}(N)$ indicates **invariance** (robustness) to population size.

Hypothesis Testing (Empirical p-value)

We test whether the **late Control** profile is consistent with neutral evolution from the **early Control** population.

- **Null hypothesis H_0:** Late Control can be generated by the neutral, symmetric model (rate p^*) starting from early Control.
- **Observed test statistic:**

$$D_{\text{obs}} = \left\| \bar{C}^{\text{late}} - \bar{C}^{\text{early}} \right\|_1 .$$

- **Null distribution:** Simulate K times under H_0 (start at early Control, run with p^*, compute $D_k = \left\| \bar{C}^{(S)}_{(k)} - \bar{C}^{\text{early}} \right\|_1$).
- **Empirical p-value:**

$$p_{\text{emp}} = \frac{1}{K} \sum_{k=1}^{K} \mathbb{I} \left(D_k \geq D_{\text{obs}} \right) .$$

Result

Control:

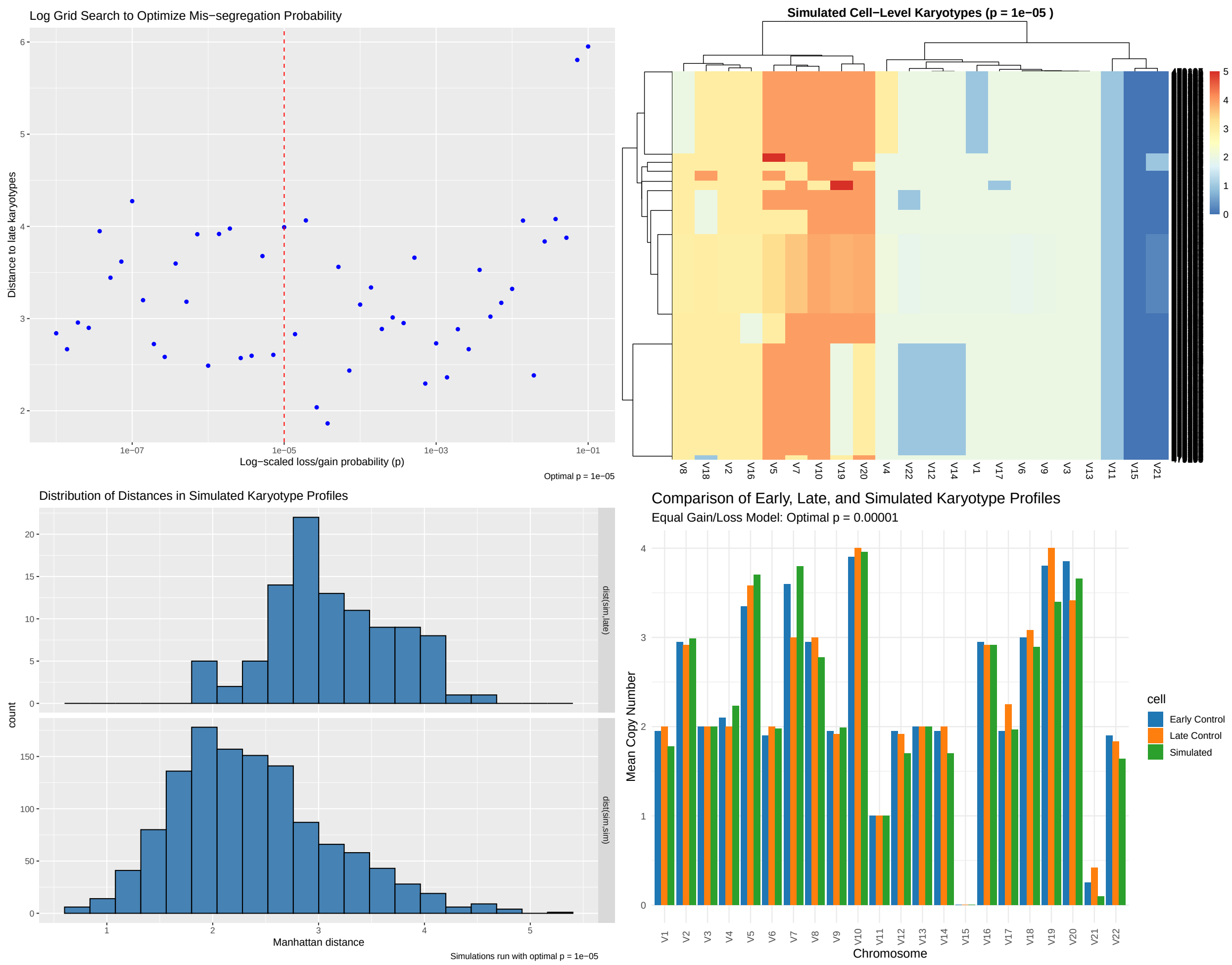

Figure 1 presents a comprehensive overview of our mechanistic simulation study modeling chromosomal evolution in the SNU-668 cell line under the Control condition. The figure integrates results from grid search optimization, simulated cell-level profiles, statistical evaluation of simulation variability, and direct comparison with empirical data.

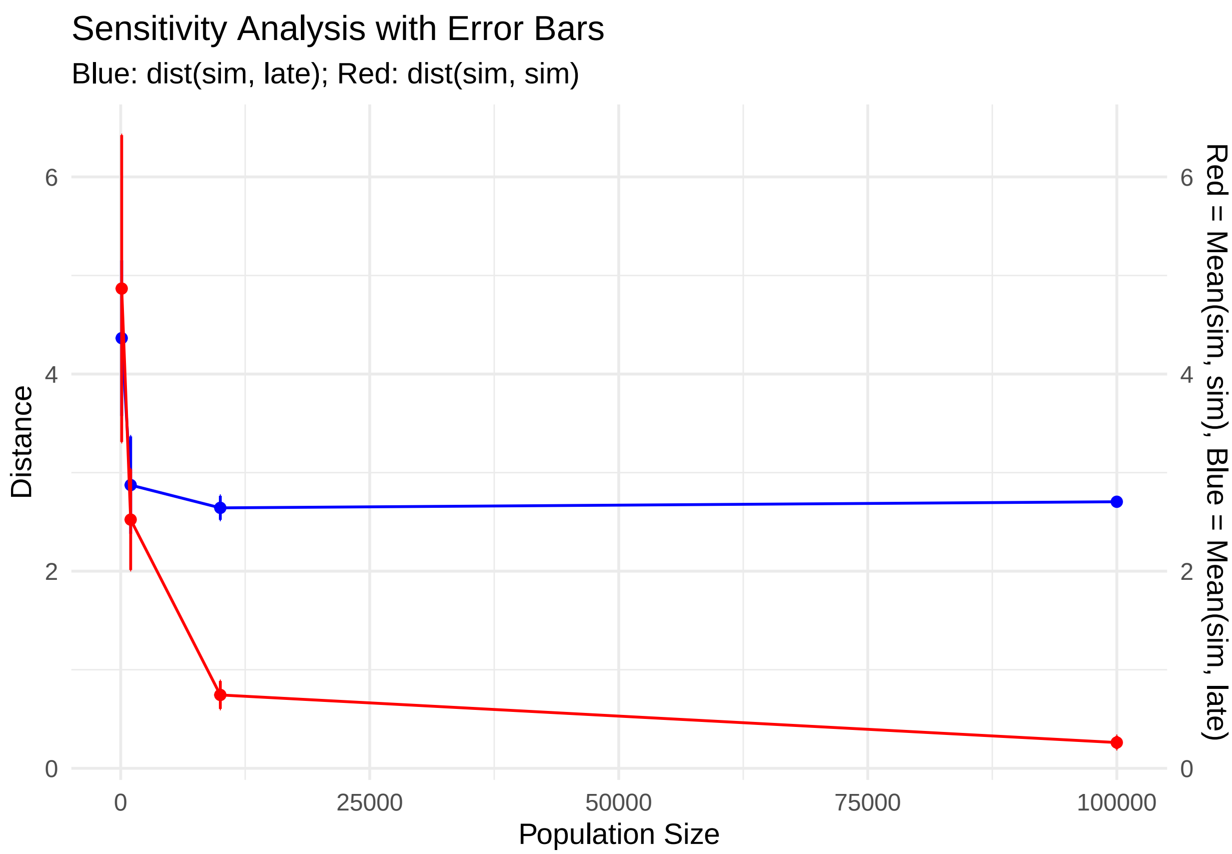

Figure 2: Sensitivity analysis showing that distance to the late-phase profile (blue) stabilizes beyond N=1,000, while inter-replicate variability (red) decreases steadily with increasing population size, indicating improved simulation consistency at larger N.

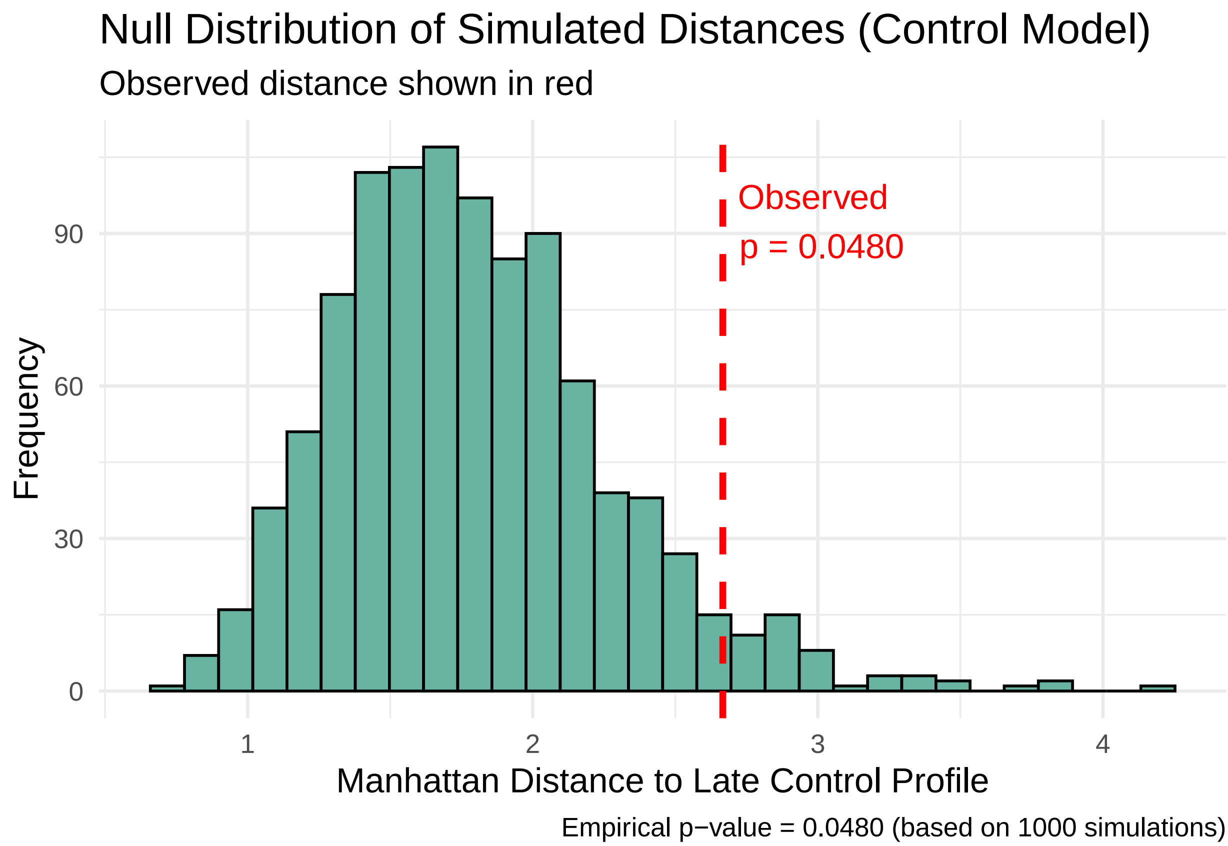

Figure 3: Null distribution of simulated distances under the control model, with the observed early-to-late control distance (red) at the 5% tail, suggesting borderline deviation from neutral evolution.

Glucose 1

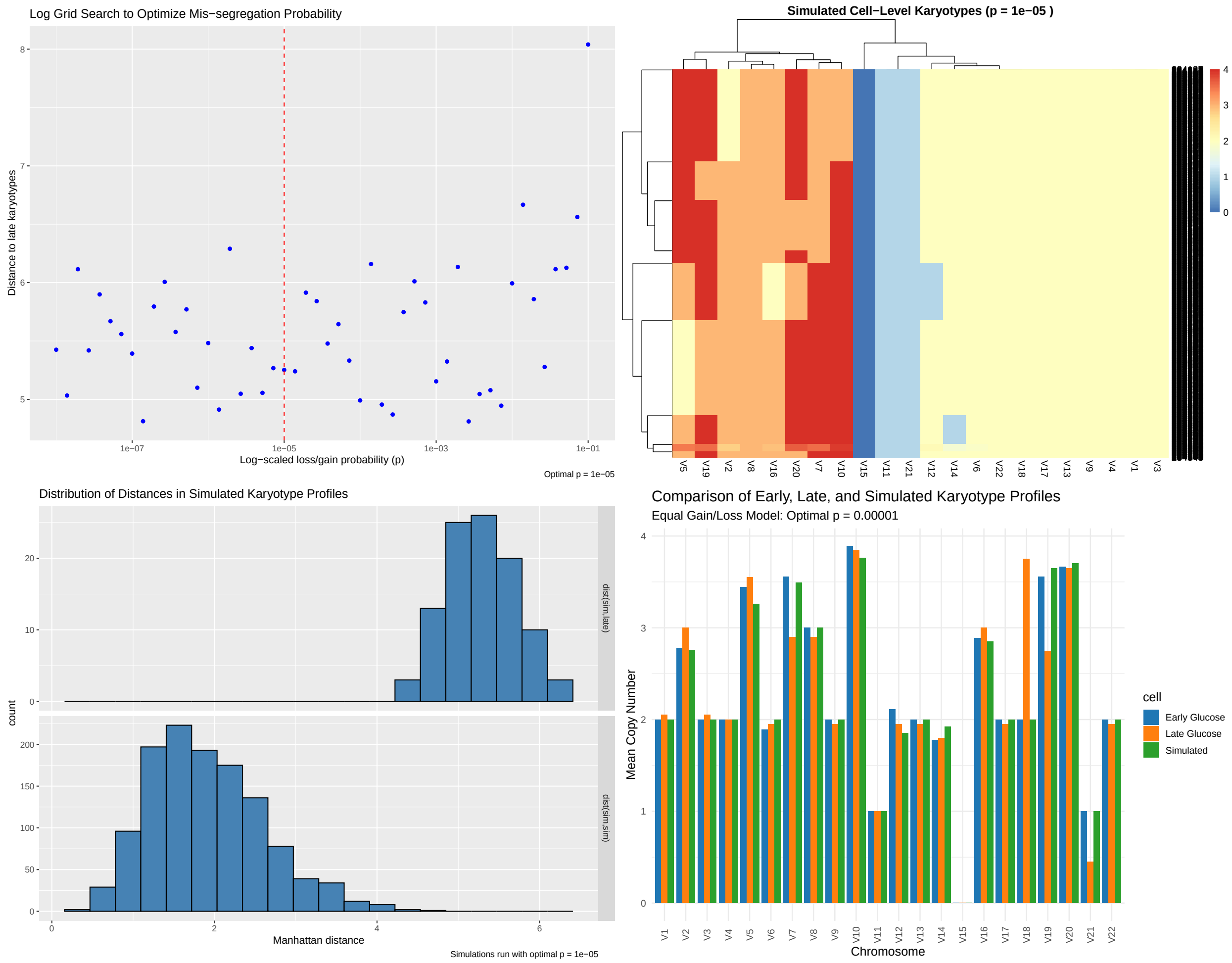

Figure 4: Simulation of karyotype evolution under the Glucose 1 condition in SNU-668 cells using a mechanistic mis-segregation model. The figure includes grid search optimization for the gain/loss probability, heatmap of simulated cell-level karyotypes, distance distributions, and comparison of early, late, and simulated profiles.

Glucose 2

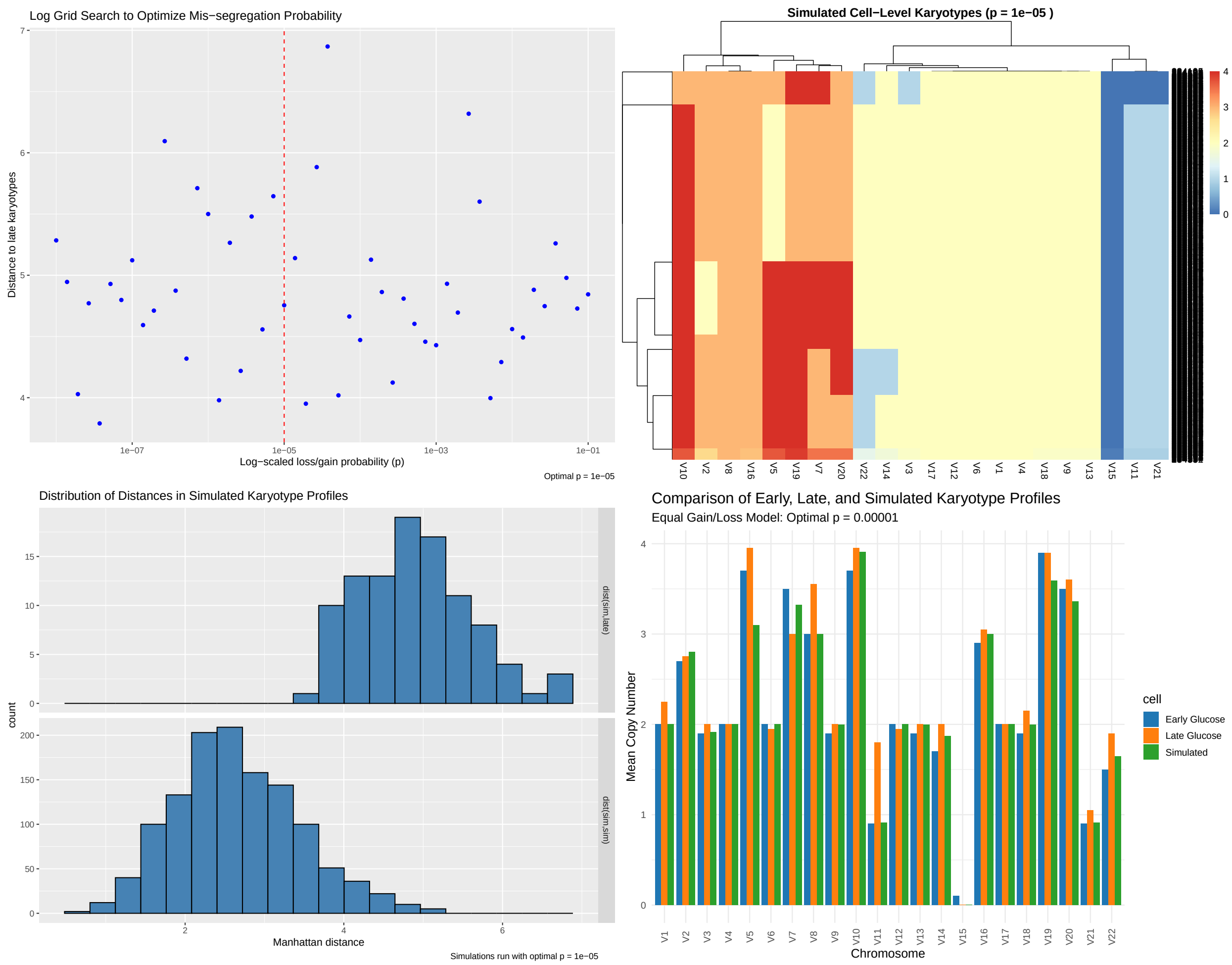

Figure 5: Overview of karyotype evolution under the Glucose 2 condition in SNU-668 cells, using a symmetric mis-segregation simulation model. The figure includes grid search optimization, heatmap of simulated karyotypes, distribution of distance metrics, and comparative copy number profiles across early, late, and simulated states.

Discussion

Our analysis demonstrates that the neutral-evolution discrete-time Markov model with symmetric chromosomal gain/loss transitions accurately describes the karyotype evolution of SNU-668 cells under **Control** conditions. Using the optimal mis-segregation probability (`p = 1e-5`), simulated late-phase profiles closely matched empirical late-phase data, with minimal variability and low Manhattan distances. This strong fit supports the conclusion that, in the absence of external stress, the control populations evolve in a manner consistent with neutral drift.

In contrast, the same neutral model failed to reproduce late-stage profiles under **Glucose 1** and **Glucose 2** conditions. Simulations showed greater divergence from the observed late-phase karyotypes, higher inter-replicate variability, and chromosome-specific discrepancies, indicating that glucose-starved populations do not follow neutral evolutionary trajectories. This mismatch suggests the presence of directional selection or stress-induced biases in chromosomal evolution under metabolic restriction, rather than the random drift seen in controls.

Robustness checks confirmed that these findings are stable across a broad range of simulated population sizes (100–100,000 cells) and that variability between simulations remained consistent under control. Hypothesis testing further showed that the late-control profile could plausibly have been generated by the neutral model (high p-value), whereas the late-glucose profiles could be rejected as model-generated with high confidence (low p-values). Together, these results provide strong evidence that while control populations evolve neutrally, glucose-starved populations deviate significantly from neutral expectations.

Conclusion

This study shows that a discrete-time Markov model with symmetric chromosomal gain/loss probabilities accurately captures the karyotype evolution of SNU-668 cells under **control** conditions. The optimal mis-segregation rate (`p = 1e-5`) produced simulated late-phase profiles that closely matched empirical observations, supporting the interpretation that control populations evolve largely through neutral drift.

In contrast, when applied to **glucose-deprived** populations (Glucose 1 and Glucose 2), the same model failed to reproduce late-stage karyotypes, producing greater divergence and variability. This clear difference between control and glucose conditions indicates that glucose-starved populations do **not** follow neutral evolutionary dynamics, likely reflecting selection pressures or stress-induced biases in chromosome segregation.

Sensitivity analyses confirmed that these findings are robust across a wide range of simulated population sizes (100–100,000 cells), and hypothesis testing provided strong statistical support: the neutral model could plausibly generate the late control profile, but late glucose profiles were rejected as model-consistent with high confidence.

Recommendations

1. **Focus on Biological Interpretation Rather Than Model Modification**
   Since the inability to fit the glucose data is itself biologically meaningful, future work should treat this as evidence of non-neutral evolution under metabolic stress, rather than attempting to adapt the neutral model to fit.

2. **Quantify Selection Dynamics**
   Follow-up studies should explore whether specific chromosomal changes confer a fitness advantage under glucose restriction, possibly using lineage tracing or time-resolved karyotype sequencing.

3. **Extend to Other Systems**
   Applying this neutral-model vs. deviation framework to other cancer cell lines and stress conditions will help determine how general the observed pattern is.

4. **Deepen Statistical Testing**
   Incorporate additional goodness-of-fit measures and likelihood-based approaches to refine p-value estimates and better characterize the statistical strength of rejecting neutrality in stressed conditions.